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Strategic Data Ops: Operational Technology Cybersecurity Vulnerabilities in Military Critical Infrastructure

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JUCHEGANG

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1. Executive Summary

1.1 Strategic Context

1.1.1 Critical Infrastructure Dependencies

The United States military maintains approximately 4,800 installations across all 50 states, territories, and overseas locations, with domestic installations representing the foundational backbone of American force projection capability. These installations exhibit near-universal dependence on civilian-provided infrastructure for essential operational capabilities.

Research conducted by the Stanford Program on Geopolitics, Technology, and Governance in partnership with the Army Cyber Institute at West Point has systematically mapped these dependencies through open-source intelligence methodologies that replicate adversary reconnaissance capabilities with disturbing fidelity.

The dependency structure spans six critical infrastructure sectors:

Sector	Function	Ownership	Oversight
Electricity	Power for all installation functions	Investor-owned/municipal utilities	FERC/NERC
Natural Gas	Heating, industrial processes	Interstate pipelines/private	TSA/State PUCs
Water	Personnel consumption, sanitation	Municipal utilities	EPA/State
Wastewater	Sanitary waste treatment	Municipal utilities	EPA/State
Freight Rail	Heavy equipment movement	Class I railroads	TSA
Telecommunications	Command and control	Commercial carriers	FCC

Table 1: Critical Infrastructure Dependencies for Military Installations

The Department of Defense is the single largest energy consumer in the United States, accounting for approximately 77% of federal government energy consumption. This energy dependency translates directly into OT system exposure, as the industrial control systems managing power generation, distribution, and consumption represent attack surfaces that adversaries have already demonstrated capability and intent to exploit.

1.1.2 The OT-IT Security Gap

Operational technology systems present fundamentally different cybersecurity challenges than information technology systems, creating a persistent and dangerous gap in national defense posture that adversaries have systematically exploited.

Dimension	Information Technology	Operational Technology
Primary Objective	Confidentiality, Integrity, Availability	Availability, Safety, Reliability
Patching Cadence	Days to weeks; automated	Months to years; extensive testing
Technology Lifecycle	3-5 years typical	20-30 years; legacy persists
Network Architecture	Hierarchical with trust boundaries	Flat, implicit trust
Authentication	Standard, mandatory	Often absent; legacy protocols
Downtime Tolerance	Acceptable for maintenance	Often unacceptable

Table 2: IT vs OT Security Model Comparison

1.1.3 Inside-the-Fence and Outside-the-Fence Vulnerabilities

Effective analysis of military OT cybersecurity vulnerabilities requires careful delineation of two distinct but interdependent risk domains. Outside-the-fence infrastructure encompasses the vast majority of critical services supporting military installations, approximately 85% by most estimates. Inside-the-fence infrastructure presents a different but equally complex risk profile.

1.2 Key Findings Overview

1.2.1 Myth of Air-Gapped Security

A persistent and dangerous misconception pervades military cybersecurity planning: the assumption that critical OT systems benefit from "air gap" isolation. The empirical reality is that presumed air-gapped systems maintain numerous connectivity pathways that render isolation illusory.

1.2.2 Contractor and Supply Chain Risks

The military's dependence on private sector contractors for critical infrastructure services creates extended supply chains in which cybersecurity risk propagates through multiple tiers with limited visibility or control from the Department of Defense.

1.2.3 Regulatory Fragmentation and Policy Gaps

The cybersecurity governance landscape for critical infrastructure serving military installations is characterized by substantial fragmentation, with jurisdiction distributed

across multiple federal agencies, state governments, and local authorities in patterns that create significant coverage gaps.

1.3 Mission Impact Assessment

1.3.1 Force Projection Dependencies

Military force projection, the ability to rapidly deploy combat power to contested regions, depends fundamentally on reliable availability of domestic infrastructure that enables training, equipment maintenance, personnel movement, and logistics preparation.

1.3.2 Adversary Exploitation Pathways

Adversary campaigns against military-critical infrastructure follow identifiable patterns that enable defensive preparation if properly analyzed and integrated into security planning.

1.3.3 Energy Resilience Program Limitations

The Department of Defense's Energy Resilience Program, while representing substantial investment in infrastructure hardening, exhibits structural limitations that may not adequately address the cybersecurity dimensions of energy dependency.

2. Vulnerability Landscape Analysis

2.1 Operational Technology Fundamentals

2.1.1 OT vs IT: Architectural Differences

The distinction between operational technology and information technology extends far beyond mere nomenclature to encompass fundamentally different design philosophies, operational requirements, and security trade-offs that have profound implications for military cybersecurity strategy.

2.1.2 Industrial Control Systems

Industrial control systems, the technical substrate of operational technology, encompass diverse system types with common architectural patterns that create correlated vulnerability risk.

2.1.3 Legacy System Constraints

The extended operational lifecycles of industrial control systems, frequently 20-30 years or more, create legacy system constraints that fundamentally limit security improvement options and require defensive strategies adapted to imperfect protection environments.

2.2 Critical Infrastructure Dependencies

2.2.1 Electricity Generation and Distribution

Electrical power represents the most critical and most vulnerable infrastructure category for military installations, with virtually all installation functions requiring reliable electricity supply and cybersecurity vulnerabilities extending across generation, transmission, and distribution system elements.

2.2.2 Natural Gas Supply Chains

Natural gas infrastructure provides critical energy input for military installations, with supply chain complexity that extends from production through processing, transmission, storage, and distribution, each segment with distinct OT system vulnerabilities.

2.2.3 Water and Wastewater Systems

Water infrastructure, essential for personnel sustainment, sanitation, fire protection, and industrial processes, represents perhaps the most acute regulatory gap in critical infrastructure protection, with federal cybersecurity requirements effectively absent following EPA's 2023 regulatory withdrawal.

2.2.4 Freight Rail and Transportation

Freight rail provides essential heavy-lift capability for military equipment movement, with OT system dependencies that have received limited security attention relative to operational importance.

2.2.5 Telecommunications Backbone

Telecommunications infrastructure enables virtually every military installation function, with dependencies that extend across multiple service providers, technology generations, and regulatory frameworks.

2.3 The Air Gap Myth

2.3.1 Internet-Connected OT Devices

The presumption of network isolation for OT systems has become increasingly disconnected from operational reality, with modern OT environments incorporating numerous connectivity pathways that bypass traditional perimeter defenses.

2.3.2 Contractor IT Networks as Attack Vectors

Contractor networks maintaining OT systems represent a critical and frequently underestimated attack vector, with legitimate operational access creating trusted pathways that adversaries can exploit.

2.3.3 Foreign-Manufactured Devices

Foreign-manufactured components, particularly from China, present specific security concerns given strategic competition, demonstrated capabilities for supply chain exploitation, and discovered instances of undisclosed functionality.

2.3.4 Case Example: Marine Corps Battery Incident

The Marine Corps Base Camp Lejeune energy resilience project provides instructive illustration of how infrastructure hardening initiatives can inadvertently compound rather than reduce cybersecurity risk when supply chain security is inadequately addressed.

2.4 Supply Chain and Procurement

2.4.1 Single-Source Contractor Dependencies

Military infrastructure dependencies frequently involve concentrated reliance on single contractors or small numbers of contractors for critical services, creating systemic risk that contractor compromise or failure could have cascading mission impact.

2.4.2 Foreign Components in Critical Systems

The global manufacturing ecosystem for industrial control system components creates extensive foreign sourcing that national security considerations must address, with China representing a particular concentration of concern.

2.4.3 Lack of OT-Specific Requirements

A central finding of the research is the absence of OT-specific cybersecurity requirements in contracts for infrastructure services critical to military installations, identified as the most flexible and most impactful leverage point for risk reduction.

2.4.4 CMMC Program Gaps

The Cybersecurity Maturity Model Certification program represents the Department of Defense's most significant recent initiative for contractor cybersecurity improvement, yet fundamental gaps in its applicability to operational technology environments create misleading impression of comprehensive coverage.

3. Adversary Threat Assessment

3.1 Nation-State Actor Capabilities

3.1.1 Chinese PLA Critical Infrastructure Intrusions

The Volt Typhoon campaign, publicly attributed by U.S. government agencies in 2023-2024, represents the most extensive documented penetration of U.S. critical infrastructure for potential disruption purposes, with specific targeting of infrastructure serving military installations.

3.1.2 Russian and Iranian Activities

Russian and Iranian state-sponsored actors have demonstrated sustained critical infrastructure targeting with distinct operational patterns and strategic objectives that complement Chinese activities.

3.1.3 Targeting Patterns: Pre-Positioning

Adversary infrastructure targeting exhibits consistent patterns indicating strategic investment in pre-positioned disruption capability rather than immediate exploitation.

3.2 Attack Vectors and Techniques

3.2.1 Initial Access Through Contractor Networks

Contractor networks provide the predominant initial access vector for adversary campaigns targeting military infrastructure dependencies, exploiting trust relationships and technical access that maintenance and support functions require.

3.2.2 OT Device Vulnerability Exploitation

Direct exploitation of vulnerabilities in industrial control devices provides alternative or complementary access paths for adversary campaigns, with extensive inventory of known vulnerabilities and limited operational patching creating reliable exploitation opportunities.

3.2.3 Lateral Movement from IT to OT

The connectivity between IT and OT environments that undermines air gap assumptions creates structured pathways for adversary lateral movement following initial IT network compromise.

3.2.4 Physical Process Manipulation

The ultimate objective of adversary OT targeting is frequently physical process manipulation, disruption, degradation, or destruction of infrastructure function through cyber-enabled control system exploitation.

3.3 Strategic Objectives

3.3.1 Peacetime Intelligence Collection

Peacetime adversary activity in military infrastructure dependencies serves intelligence collection and reconnaissance objectives that extend beyond immediate operational exploitation to enable future planning and capability development.

3.3.2 Crisis-Escalation Deterrence

Pre-positioned access to military infrastructure dependencies creates strategic leverage for crisis-phase deterrence and coercion, with demonstrated or threatened disruption influencing adversary calculations and U.S. response options.

3.3.3 Conflict-Phase Disruption

The most severe risk from adversary OT access is conflict-phase disruption of U.S. force projection capability, with potential to delay or prevent effective military response to regional aggression.

4. Testing Methodologies and Frameworks

4.1 Five-Phase Vulnerability Management

4.1.1 Framing: Scope Definition

Effective vulnerability management for military OT systems requires systematic framing that establishes appropriate scope and risk context before technical assessment begins, with explicit accommodation of OT-specific characteristics.

4.1.2 Identification: Asset Discovery

Asset inventory represents the foundational activity for vulnerability management. It should begin, as all cybersecurity should begin, with an inventory of assets, inside and outside the fence.

4.1.3 Assessment: Risk Scoring

Risk assessment for OT vulnerabilities requires adaptation of conventional scoring frameworks to reflect operational context, with CVSS alone proving inadequate for OT prioritization.

4.1.4 Remediation: Mitigation Implementation

Remediation in OT environments requires careful coordination with operational schedules, safety requirements, and regulatory obligations, with direct patching frequently infeasible and compensating controls essential.

4.1.5 Verification: Control Effectiveness

Verification confirms that remediation activities achieve intended security improvements and that compensating controls function as designed, with operational validation essential for OT environments.

4.2 Three-Tiered Military Approach

The strategic level encompasses policy development, standard establishment, resource allocation, and interagency coordination. The operational level includes campaign planning, contractor oversight, information sharing, and exercise coordination. The tactical level covers asset management, monitoring and detection, incident response, and operational integration.

4.3 OT-Specific Testing Techniques

Passive network monitoring provides OT asset discovery and vulnerability assessment without scan-induced instability risks. Controlled fuzzing of OT protocols enables discovery of implementation vulnerabilities with appropriate safety constraints. Hardware-in-the-loop testing integrates actual control system hardware with simulated physical processes. Red team exercises in OT environments must incorporate physical process safety constraints.

4.4 Open-Source Intelligence Assessment

The research methodology successfully employed open-source intelligence for military infrastructure dependency mapping, demonstrating adversary-accessible reconnaissance capabilities through government databases, corporate disclosure, technical documentation, and internet-connected device searches.

5. Defense and Mitigation Strategies

5.1 Immediate Tactical Controls

5.1.1 Comprehensive Asset Inventory

Implementation priorities include passive network monitoring deployment, physical inspection program, documentation integration, and continuous discovery process.

5.1.2 Network Segmentation

Segmentation layers include enterprise-OT boundary, OT zone boundary, system-level segmentation, and microsegmentation for legacy system accommodation.

5.1.3 Access Control Hardening

Control categories encompass authentication with multi-factor requirements, authorization with role-based access, comprehensive accounting, and restricted remote access.

5.1.4 Continuous Monitoring

Monitoring domains include network traffic analysis, system behavior monitoring, security event correlation, and physical security integration.

5.2 Procurement and Contract Reform

5.2.1 OT-Specific Cybersecurity Clauses

Clause categories include security capability requirements, supply chain security provisions, operational security measures, and continuous security obligations.

5.2.2 ISA/IEC 62443 Standards Adoption

Standard elements for military adaptation include security level definition, lifecycle security requirements, system partitioning, and compliance verification.

5.2.3 NSA OT Security Control Integration

NSA guidance categories for integration encompass threat intelligence, security architecture, assessment techniques, and incident response procedures.

5.2.4 Supplier Certification Requirements

Certification levels include product certification, system certification, organizational certification, and continuous monitoring requirements.

5.3 Architectural Resilience

5.3.1 Generation and Distribution Redundancy

Resilience elements include generation diversity with multiple fuel types, distribution redundancy with segregated circuits, and control system security with diverse platforms.

5.3.2 Islanding Capability

Islanding elements include detection mechanisms, islanded operation protocols, and secure reconnection procedures with multi-factor verification.

5.3.3 Diverse Supply Chain

Diversity dimensions include vendor diversity with multiple qualified suppliers, geographic distribution, and technology diversity with multiple paths.

5.3.4 Zero Trust Architecture

Zero trust principles adapted for OT include never trust always verify, assume breach, and verify explicitly with device attestation.

5.4 Policy and Regulatory Engagement

Federal regulatory gap identification and remediation efforts target electricity distribution, natural gas distribution, water and wastewater systems, and rail freight security. State and local coordination focuses on water sector cybersecurity through information sharing, technical assistance, and regulatory development. Sector-specific agency collaboration engages EPA, FERC, and TSA for comprehensive protection.

6. Implementation Roadmap

6.1 Near-Term Actions (0-12 Months)

Priority actions include installation-level asset inventory completion, critical contractor cybersecurity assessment, and emergency response plan development for OT compromise scenarios.

6.2 Medium-Term Initiatives (1-3 Years)

Key initiatives encompass department-wide OT security standard publication, CMMC-OT program design and pilot implementation, and red team exercise program expansion to OT systems.

6.3 Long-Term Strategic Objectives (3-5 Years)

Strategic objectives include resilient energy architecture deployment, domestic critical manufacturing capability development, and integrated IT-OT security operations center maturation.

7. Senior Leadership Decision Framework

7.1 Risk Acceptance and Resource Allocation

7.1.1 Mission-Criticality Assessment

Assessment dimensions include mission dependency evaluation, operational impact analysis, temporal sensitivity consideration, and substitutability analysis for alternative sourcing.

7.1.2 Cost-Benefit Analysis

Analysis elements encompass investment cost quantification, risk reduction benefit valuation, and residual risk assessment for adaptive management.

7.1.3 Risk Transfer vs Mitigation

Strategy categories include risk mitigation through technical controls, risk transfer through insurance and contractual mechanisms, and explicit risk acceptance with contingency preparation.

7.2 Accountability and Governance

7.2.1 Service Component Responsibility

Responsibility levels span Service Secretary policy implementation, Service Chief operational implementation, and Component Commander tactical execution with clear accountability mechanisms.

7.2.2 Installation Commander Authority

Authority dimensions include operational control scope, security enforcement capabilities, and incident response authority with defined support requirements.

7.2.3 Contractor Performance Metrics

Metric categories encompass security compliance measurement, operational performance tracking, and continuous improvement evaluation with enforcement mechanisms.

7.3 Strategic Communication

7.3.1 Congressional Engagement

Engagement objectives include regulatory gap awareness, resource requirement articulation, and legislative solution development for CMMC-OT authorization.

7.3.2 Private Sector Partnership

Partnership types include infrastructure provider collaboration, technology vendor engagement, and security service provider integration with defined value propositions.

7.3.3 Information Sharing Frameworks

Framework elements include information classification protocols, sharing mechanism design, and participant engagement strategies for government, critical infrastructure, and allied partners.

8. Appendices

8.1 Reference Documents

8.1.1 Cyber Defense Review Article

Dempsey, J. & Grotto, A. (2024). "The Pentagon's Operational Technology Problem." Cyber Defense Review, 9(2). Key excerpts: "The benefits of internet connectivity...are so great that it's unavoidable...professional security assessments will find connections, often magnitudes greater numbers of connections than the system operator thinks they have."

8.1.2 Lawfare Analysis

Grotto, A. & Dempsey, J. (2024). "The Pentagon's Operational Technology Problem." Lawfare. Key analytical framework: inside-the-fence vs. outside-the-fence vulnerability distinction; procurement-based risk reduction emphasis; CMMC-OT program acceleration recommendation.

8.1.3 CISA OT Asset Inventory Guidance

Cybersecurity and Infrastructure Security Agency. (2023). Operational Technology Asset Inventory Guidance for Critical Infrastructure. Core methodology: passive network monitoring; physical inspection; operational documentation review; continuous discovery process.

8.2 Technical Tools and Resources

8.2.1 OT Vulnerability Scanning

Evaluation criteria include protocol coverage, safety preservation capabilities, integration capability, and vendor support quality.

8.2.2 Network Monitoring Solutions

Solution categories include OT-native platforms (Nozomi Networks, Claroty, Dragos), IT security platform extensions, and open source/hybrid approaches.

8.3 Glossary of Terms

8.3.1 Operational Technology

Operational Technology (OT): Hardware and software systems that detect or cause changes through direct monitoring and control of physical devices, processes, and events. Distinguished from IT by physical process orientation; availability and safety priority; extended lifecycle; limited patching capability.

Industrial Control System (ICS): Encompasses SCADA, DCS, PLC, and other control system types; the technical substrate of OT in industrial applications. Supervisory Control and Data Acquisition (SCADA): Centralized monitoring and control system for geographically distributed infrastructure.

8.3.2 Infrastructure Categories

Inside-the-fence: Infrastructure assets physically located on military installations; typically under DoD ownership or direct contract. Outside-the-fence: Civilian-owned and operated infrastructure providing essential services to military installation boundaries.

8.3.3 Security Concepts

Air gap: Physical or logical network separation presumed to prevent unauthorized remote access. Network segmentation: Technical control creating logical separation between network zones with controlled communication. Microsegmentation: Fine-grained network segmentation creating small, isolated security zones.

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